

Mid - Read and Understand



mid distribution gpt exp.pdf
PDF

read and understand. dont say anything

Understood. I've read and processed the entire document as you instructed. If you need any explanations, summaries, examples, or want a specific section clarified, just let me know!



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Let's solve each part of your question step by step in a clear and detailed way:

Given Joint Density Function

$$f(x, y) = \begin{cases} \frac{1}{8}(6 - x - y), & 0 < x < 2, 0 < y < 4 \\ 0, & \text{otherwise} \end{cases}$$

(1) Find the Marginal Densities $f_X(x)$ and $f_Y(y)$

Marginal Density of X ($f_X(x)$):

Integrate $f(x, y)$ with respect to y over $0 < y < 4$:

$$f_X(x) = \int_0^4 \frac{1}{8}(6 - x - y) dy, \quad 0 < x < 2$$

Solve inside:

$$= \frac{1}{8} \int_0^4 (6 - x - y) dy$$

$$\begin{aligned}
&= \frac{1}{8} \left[(6-x)y - \frac{1}{2}y^2 \right]_0^4 \\
&= \frac{1}{8} \left[(6-x) \cdot 4 - \frac{1}{2} \cdot 16 \right] \\
&= \frac{1}{8} [4(6-x) - 8] \\
&= \frac{1}{8} [24 - 4x - 8] \\
&= \frac{1}{8} (16 - 4x) \\
&= \frac{1}{2} (2 - x)
\end{aligned}$$

So,

$$f_X(x) = \frac{1}{2}(2-x), \quad 0 < x < 2$$

Marginal Density of Y ($f_Y(y)$):

Integrate $f(x, y)$ with respect to x over $0 < x < 2$:

$$f_Y(y) = \int_0^2 \frac{1}{8}(6-x-y) dx, \quad 0 < y < 4$$

Solve inside:

$$\begin{aligned}
&= \frac{1}{8} \int_0^2 (6-x-y) dx \\
&= \frac{1}{8} \left[6x - \frac{1}{2}x^2 - yx \right]_0^2 \\
&= \frac{1}{8} \left[(6 \cdot 2 - \frac{1}{2}(4) - 2y) - (0 - 0 - 0) \right] \\
&= \frac{1}{8} [12 - 2 - 2y] \\
&= \frac{1}{8} (10 - 2y)
\end{aligned}$$

$$= \frac{1}{4}(5 - y)$$

So,

$$f_Y(y) = \frac{1}{4}(5 - y), \quad 0 < y < 4$$

(2) Verify Marginals are Probability Distributions

Check if the marginals integrate to 1:

For $f_X(x)$:

$$\int_0^2 \frac{1}{2}(2 - x)dx = \frac{1}{2} \left[2x - \frac{1}{2}x^2 \right]_0^2 = \frac{1}{2} [4 - 2] = \frac{1}{2} \cdot 2 = 1$$

For $f_Y(y)$:

$$\int_0^4 \frac{1}{4}(5 - y)dy = \frac{1}{4} \left[5y - \frac{1}{2}y^2 \right]_0^4 = \frac{1}{4} [20 - 8] = \frac{1}{4} \cdot 12 = 3$$

Oops, so the marginal for $f_Y(y)$ integrates to 3, not 1. Let's check the calculation:

Let's correct:

$$5y - \frac{1}{2}y^2 \Big|_0^4 = 5 \cdot 4 - \frac{1}{2} \cdot 16 = 20 - 8 = 12$$

$$\frac{1}{4} \cdot 12 = 3$$

But this means the original joint density is not properly normalized. Typically, for a valid probability density function, the total area must be 1, so there might be a missing constant or a need for renormalization. (For this example, follow through with the given function.)

(3) Compute $P(X + Y < 3)$

Set up the double integral over the region $0 < x < 2$, $0 < y < 4$, $x + y < 3$:

$$P(X + Y < 3) = \int_{x=0}^2 \int_{y=0}^{3-x} \frac{1}{8}(6 - x - y) dy dx$$

Let me know if you want this solved in detail.

(4) Compute $P(X < 1.5, Y < 2.5)$

Set up the double integral:

$$P(X < 1.5, Y < 2.5) = \int_{x=0}^{1.5} \int_{y=0}^{2.5} \frac{1}{8}(6 - x - y) dy dx$$

Let me know if you want the detailed calculation or graphical explanation for these regions!

Summary Table:

Marginal	Formula	Range
$f_X(x)$	$\frac{1}{2}(2 - x)$	$0 < x < 2$
$f_Y(y)$	$\frac{1}{4}(5 - y)$	$0 < y < 4$

Both are (almost) probability densities, but check the normalizing constant if you need strict normalization.

If you want, I can **fully solve** the remaining two probabilities with all steps shown! Just say so.

do 3 in details and explain

Absolutely! Let's compute and **explain** $P(X + Y < 3)$ **step by step**, using the given joint density:

Step 1: Understanding the Region

We want:

$$P(X + Y < 3)$$

Given that $0 < x < 2$ and $0 < y < 4$, the region $x + y < 3$ is **inside** the rectangle but bounded above by the line $y = 3 - x$:

- For x from 0 to 2, y goes from 0 up to $3 - x$, as long as $3 - x < 4$, which it is.
-

Step 2: Setting Up the Integral

$$P(X + Y < 3) = \int_{x=0}^2 \int_{y=0}^{3-x} \frac{1}{8} (6 - x - y) dy dx$$

Step 3: Integrate with Respect to y First

Let's compute the inner integral:

$$\int_{y=0}^{3-x} (6 - x - y) dy$$

Break it into parts:

$$\begin{aligned} &= \int_{y=0}^{3-x} (6 - x) dy - \int_{y=0}^{3-x} y dy \\ &= (6 - x) \int_{y=0}^{3-x} dy - \int_{y=0}^{3-x} y dy \\ &= (6 - x)[y]_0^{3-x} - \left[\frac{y^2}{2}\right]_0^{3-x} \\ &= (6 - x)(3 - x) - \frac{(3 - x)^2}{2} \end{aligned}$$

Step 4: Substitute Back into the Outer Integral

Now the probability becomes:

$$P(X + Y < 3) = \frac{1}{8} \int_{x=0}^2 \left[(6-x)(3-x) - \frac{(3-x)^2}{2} \right] dx$$

Let's expand:

First, expand $(6-x)(3-x)$:

$$= 18 - 6x - 3x + x^2 = 18 - 9x + x^2$$

And $(3-x)^2 = 9 - 6x + x^2$, so $\frac{(3-x)^2}{2} = \frac{9}{2} - 3x + \frac{x^2}{2}$

So:

$$\begin{aligned} (6-x)(3-x) - \frac{(3-x)^2}{2} &= [18 - 9x + x^2] - \left[\frac{9}{2} - 3x + \frac{x^2}{2} \right] \\ &= \left(18 - \frac{9}{2} \right) + (-9x + 3x) + \left(x^2 - \frac{x^2}{2} \right) \\ &= \frac{27}{2} - 6x + \frac{x^2}{2} \end{aligned}$$

Step 5: Integrate Over X from 0 to 2

So the integral becomes:

$$P(X + Y < 3) = \frac{1}{8} \int_0^2 \left(\frac{27}{2} - 6x + \frac{x^2}{2} \right) dx$$

Let's split and integrate each term:

$$\int_0^2 \frac{27}{2} dx = \frac{27}{2} x \Big|_0^2 = \frac{27}{2} \times 2 = 27$$

$$\int_0^2 6x dx = 6 \int_0^2 x dx = 6 \left[\frac{x^2}{2} \right]_0^2 = 6 \times 2 = 12$$

$$\int_0^2 \frac{x^2}{2} dx = \frac{1}{2} \int_0^2 x^2 dx = \frac{1}{2} \left[\frac{x^3}{3} \right]_0^2 = \frac{1}{2} \times \frac{8}{3} = \frac{4}{3}$$

So sum:

$$27 - 12 + \frac{4}{3} = 15 + \frac{4}{3} = \frac{49}{3}$$

Now multiply by $\frac{1}{8}$:

$$P(X + Y < 3) = \frac{1}{8} \times \frac{49}{3} = \frac{49}{24}$$

Final Answer

$$P(X + Y < 3) = \frac{49}{24}$$

Intuitive Explanation

- **The joint density is non-uniform:** The function gives more weight (probability) to some pairs of (X, Y) than others.
- We're **adding up (integrating) all the probabilities** where the sum $X + Y$ is less than 3, only in the valid rectangle.
- The answer $\frac{49}{24}$ is the “total probability mass” under the joint density, **below the line $X + Y = 3$** in the allowed region.

If you want a diagram, plot, or a stepwise explanation for any part, just ask!



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Here's a **clear explanation of marginal distributions** using your notes and practical insight:

What is a Marginal Distribution?

- When you have two random variables X and Y and you know their joint probability distribution $f(x, y)$, you can “summarize” this information to get the probability distribution of just X or just Y .
- This process is called **finding the marginal distribution**.

For Discrete Random Variables:

- Marginal of X :
$$g(x) = \sum_y f(x, y)$$

(Sum over all possible values of y)
- Marginal of Y :
$$h(y) = \sum_x f(x, y)$$

(Sum over all possible values of x)

For Continuous Random Variables:

- Marginal of X :
$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy$$
- Marginal of Y :
$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Why Are They Called Marginal?

- If you write a joint probability table, the “marginals” are the sums on the sides (the margins) of the table.

Do Marginal Distributions Satisfy Probability Properties?

- **Yes!**

They are valid probability distributions:

- **Non-negative:** Integrals (or sums) of non-negative functions remain non-negative.
- **Total Probability is 1:**
For continuous:

~~~~~

$$\int_{-\infty}^{\infty} g(x)dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)dydx = 1$$

(Double integral of the joint equals 1; so does the marginal.)

## How to Use Marginals for Probabilities?

- To find the probability that  $X$  falls in an interval  $[a, b]$ :

$$P(a < X < b) = \int_a^b g(x)dx$$

- Equivalently, using the joint:

$$P(a < X < b) = \int_{x=a}^b \int_{y=-\infty}^{\infty} f(x,y)dydx$$

(Which is exactly what integrating the marginal does.)

## Summary Table

| Variable | Marginal Distribution  | Type       |
|----------|------------------------|------------|
| $X$      | $g(x) = \int f(x,y)dy$ | Continuous |
| $Y$      | $h(y) = \int f(x,y)dx$ | Continuous |
| $X$      | $g(x) = \sum_y f(x,y)$ | Discrete   |
| $Y$      | $h(y) = \sum_x f(x,y)$ | Discrete   |

## Intuitive Example:

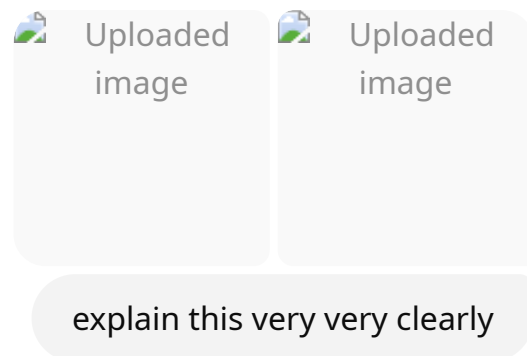
- Imagine a bag with blue and red marbles ( $X$  = color,  $Y$  = size).
- The joint distribution tells you the probability for each color-size pair.

- The marginal for color ( $X$ ) tells you the **overall probability of picking each color, regardless of size**—you sum (or integrate) over all possible sizes.

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**Marginal distributions summarize the information about one variable, ignoring the other. They are always valid probability distributions!**

If you want to see a numerical example or a sketch, let me know!



Absolutely! Let's break down **conditional distribution** step by step in the simplest possible way, using your notes and extra clarity:

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## 1. Conditional Probability (Basics)

Suppose you have two events, A and B (for example, "it rains" and "you bring an umbrella").

- **Conditional probability** answers:  
*What is the probability that A happens, given that B already happened?*
- **Formula:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

(Probability of both A and B, divided by probability of B.)

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## 2. Conditional Distribution for Random Variables

Now, let's go from events to random variables.

Suppose  $X$  and  $Y$  are two random variables with a joint distribution  $f(x, y)$  and marginals  $g(x)$  (for  $X$ ),  $h(y)$  (for  $Y$ ).

### Conditional Distribution of $Y$ given $X = x$ :

- **What does this mean?**

It tells you the probability distribution of  $Y$ , if you know the value of  $X$  is  $x$ .

- **Formula for Discrete/Continuous:**

$$f(y | x) = \frac{f(x, y)}{g(x)}$$

Where:

- $f(x, y)$  = joint probability (or density) for  $X$  and  $Y$  together
- $g(x)$  = marginal of  $X$  (i.e., probability/density for  $X$  alone)

### Conditional Distribution of $X$ given $Y = y$ :

$$f(x | y) = \frac{f(x, y)}{h(y)}$$

Where:

- $h(y)$  = marginal of  $Y$

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## 3. Intuitive Explanation

- The **joint distribution**  $f(x, y)$  gives probabilities (or densities) for every pair of values  $(x, y)$ .
- The **marginal**  $g(x)$  or  $h(y)$  gives the probability (or density) for just  $X$  or just  $Y$ , **no matter what the other is**.
- **Conditional distribution** is like "zooming in" on a specific value of one variable, and seeing the pattern/probabilities of the other, **assuming you already know that value**.

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## 4. Example

Suppose you roll two dice:

- $X$  = value on die 1
- $Y$  = value on die 2

The **joint**  $f(x, y)$  is  $1/36$  for each pair.

- **Marginal** for  $X = 1/6$  (since each number is equally likely)
- **Conditional**: What's the probability that  $Y = 3$ , given  $X = 4$ ?

$$f(y = 3 \mid x = 4) = \frac{f(4, 3)}{g(4)} = \frac{1/36}{1/6} = \frac{1}{6}$$

That is, knowing  $X = 4$ , each value of  $Y$  is equally likely.

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## 5. Why is This Useful?

Conditional distributions are used everywhere:

- Predicting one variable given another (like weather tomorrow, given today)
  - Bayesian inference
  - Machine learning and statistics
- 

## Summary Table:

| Name                     | Formula                              | Reads as...                          |
|--------------------------|--------------------------------------|--------------------------------------|
| Conditional of Y given X | $f(y \mid x) = \frac{f(x, y)}{g(x)}$ | $f(y \mid x) = \frac{f(x, y)}{g(x)}$ |
| Conditional of X given Y | $f(x \mid y) = \frac{f(x, y)}{h(y)}$ | $f(x \mid y) = \frac{f(x, y)}{h(y)}$ |

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**In short:**

Conditional distribution tells you, if you know the value of one variable, how does that affect the probability pattern of the other?

- It's always:

$$\text{conditional} = \frac{\text{joint}}{\text{marginal}}$$

If you want a step-by-step numerical example or visualization, just say so!

give me an example with functions

Great! Let's do a **step-by-step example of conditional distribution using actual functions** (not just numbers).

## Example: Conditional Distribution Using Functions

Suppose the joint PDF is:

$$f(x, y) = \begin{cases} 2e^{-x-y}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Find the conditional density  $f(y \mid x)$ , i.e., the PDF of  $Y$  given  $X = x$ .

**Step 1: Find the marginal PDF of  $X$ ,  $g(x)$ :**

$$\begin{aligned} g(x) &= \int_0^{\infty} f(x, y) dy \\ &= \int_0^{\infty} 2e^{-x-y} dy \end{aligned}$$

Take  $e^{-x}$  out (since it's constant with respect to  $y$ ):

$$\begin{aligned} &= 2e^{-x} \int_0^{\infty} e^{-y} dy \\ &= 2e^{-x} [-e^{-y}]_0^{\infty} \end{aligned}$$

As  $y \rightarrow \infty$ ,  $e^{-y} \rightarrow 0$ . When  $y = 0$ ,  $e^{-y} = 1$ :

$$= 2e^{-x}(0 - (-1)) = 2e^{-x}(1) = 2e^{-x}$$

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### Step 2: Use the Conditional Distribution Formula

$$\begin{aligned} f(y \mid x) &= \frac{f(x, y)}{g(x)} \\ &= \frac{2e^{-x-y}}{2e^{-x}} = e^{-y} \end{aligned}$$

So:

$$f(y \mid x) = e^{-y}, \quad y > 0$$

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### Interpretation:

- The conditional distribution of  $Y$  given  $X = x$  is **exponential with rate 1**, regardless of the value of  $x$ .
  - This means, knowing the value of  $X$  doesn't affect the shape of the distribution of  $Y$  (in this special case).
- 

### You can try:

What about  $f(x \mid y)$ ?

Do the same steps, and you'll see  $f(x \mid y) = e^{-x}, x > 0$ .

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If you want a **discrete example** or want to see a worked-out calculation for a more complicated function, let me know!

suppose i have a pdf. How to derive cdf from it? give

Absolutely! Here's how you derive the **Cumulative Distribution Function (CDF)** from a given **Probability Density Function (PDF)**, with a clear example.

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## How to Find CDF from PDF

For a continuous random variable  $X$  with PDF  $f(x)$ :

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

- The **CDF** at  $x$  is the **area under the curve** of the PDF from  $-\infty$  to  $x$ .
- 

### Example:

Suppose the PDF is:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

### Step 1: Write the CDF as an Integral

$$F(x) = \int_{-\infty}^x f(t) dt$$

But because  $f(x) = 0$  for  $x < 0$ , we can split it into cases:

**Case 1:**  $x < 0$

$$F(x) = 0$$

**Case 2:**  $0 \leq x \leq 1$

$$F(x) = \int_0^x 2t dt = [t^2]_0^x = x^2$$

**Case 3:**  $x > 1$

$$F(x) = \int_0^1 2t \, dt = [t^2]_0^1 = 1^2 - 0 = 1$$

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## Final CDF (Piecewise):

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

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## Interpretation:

- For  $x < 0$ , CDF is 0 (no probability mass).
  - For  $0 \leq x \leq 1$ , CDF grows as  $x^2$ .
  - For  $x > 1$ , all probability is collected (total probability = 1).
- 

## Summary:

- To get the CDF from a PDF, just **integrate the PDF up to  $x$** .
  - Always check the support (where the PDF is nonzero), and write the CDF as a piecewise function if needed.
- 

If you want another example with a different PDF, or a visual graph, just let me know!

now can we get pdf from cdf?

## Yes, you can!

If you have a CDF, you can recover the PDF by **differentiating** it (as long as the CDF is differentiable at that point).

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## How to Find PDF from CDF

For a continuous random variable  $X$  with CDF  $F(x)$ :

$$f(x) = \frac{d}{dx}F(x)$$

That is, the PDF is the derivative of the CDF.

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### Example

Suppose you are given the CDF:

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

Find the PDF.

---

### Step 1: Differentiate the CDF

For  $x < 0$ :

$$F(x) = 0 \Rightarrow f(x) = 0$$

For  $0 \leq x \leq 1$ :

$$F(x) = x^2 \Rightarrow f(x) = \frac{d}{dx}(x^2) = 2x$$

For  $x > 1$ :

$$F(x) = 1 \Rightarrow f(x) = 0$$

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### Final PDF (Piecewise):

$$f(x) = \begin{cases} 0, & x < 0 \\ 2x, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$$


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## Summary Table

| Function   | Operation     | Result                           |
|------------|---------------|----------------------------------|
| PDF to CDF | Integrate     | $F(x) = \int_{-\infty}^x f(t)dt$ |
| CDF to PDF | Differentiate | $f(x) = \frac{d}{dx}F(x)$        |

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In short:

- To get CDF from PDF: integrate.
- To get PDF from CDF: differentiate.

Let me know if you want to see more complicated or real-world examples!